

Direction 2: a compactness program for convex Schiffer counterexamples

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Scope

Direction 2 of `ideas-schiffer.tex` is the *unweighted* compactness program for the planar Schiffer conjecture. The previous notes in this folder (`direction2-barrel-limit`, `direction2-heat-extension-computa`, `direction2-parabolic-to-classical`, `direction2-stationary-weighted-perturbation`) explored a log-concave extension of the problem; that line eventually showed the parabolic angle is too weak to close the conjecture (Theorem 1 of `direction2-parabolic-to-classical` reduces it to the classical statement) and is therefore a detour relative to Direction 2 as written. This note restarts cleanly: it sets up the compactness program for genuine planar convex Schiffer counterexamples and identifies precisely where the tools from arXiv:2412.06344 enter as enabling machinery.

1 The compactness program in precise form

Conjecture 1 (Schiffer for planar convex domains). *If $\Omega \subset \mathbb{R}^2$ is a bounded smooth strictly convex domain admitting a nonzero solution of*

$$\Delta u + \lambda u = 0 \text{ in } \Omega, \quad \partial_\nu u = 0 \text{ on } \partial\Omega, \quad u|_{\partial\Omega} = c \in \mathbb{R}, \quad (1)$$

for some $\lambda > 0$, then Ω is a disc.

The compactness strategy of Direction 2: assume the negation, take a sequence $\{(\Omega_j, u_j, \lambda_j)\}$ of normalized non-disc convex counterexamples, extract a limit by Blaschke selection, pass the overdetermined data to the limit, and derive a contradiction. The program splits cleanly into four steps.

Step 2 (Setup and normalization). *Choose normalizations on the sequence so that Blaschke selection gives a non-degenerate convex limit Ω_∞ , and so that the eigenvalues λ_j are bounded in a compact subinterval of $(0, \infty)$.*

Step 3 (Spectral compactness). *Show that the normalized eigenfunctions u_j , after possibly extracting subsequences, converge in a strong enough topology (uniformly with bounded derivatives) to a limit u_∞ on Ω_∞ , and that $\lambda_j \rightarrow \lambda_\infty > 0$ with $u_\infty \not\equiv 0$.*

Step 4 (Passing the overdetermination). *Show that u_∞ satisfies (1) on Ω_∞ , including both overdetermined conditions $\partial_\nu u_\infty = 0$ and $u_\infty|_{\partial\Omega_\infty} = c_\infty$.*

Step 5 (Disposing of the disc limit). *Either rule out $\Omega_\infty = \text{disc}$ by a local rigidity theorem near the disc, or rule out a non-disc Ω_∞ by an extremality argument applied to a minimizing sequence.*

2 Step 1: normalizations

Three families of normalization are compatible with Schiffer's overdetermined problem, since (1) is invariant under $(\Omega, u, \lambda) \mapsto (\tau\Omega, \tau \circ u, \lambda)$ for rigid motions τ and under $(\Omega, u, \lambda) \mapsto (r\Omega, u(\cdot/r), \lambda/r^2)$ for dilations. Centre at 0 to remove translations; rotate so that the diameter axis is horizontal to remove rotations. To remove the dilation freedom, fix one scalar geometric quantity.

- **Diameter normalization:** $\text{diam}(\Omega_j) = 1$. Then $\Omega_j \subset B_1(0)$ and Blaschke gives a convex Hausdorff limit Ω_∞ . The drawback is that Ω_j can degenerate to a line segment (zero width).
- **Area normalization:** $|\Omega_j| = \pi$ (the area of a unit disc). Now Ω_j can have arbitrarily large diameter (long thin rectangle), so an additional bound is required to apply Blaschke.
- **Combined normalization:** $|\Omega_j| = \pi$ and $\text{diam}(\Omega_j) \leq D_*$, with D_* a constant chosen below.

Lemma 6 (Diameter bound from a fixed eigenvalue). *If $\lambda_j \leq \Lambda$ for all j and $|\Omega_j| = \pi$, then there is $D_* = D_*(\Lambda)$ such that $\text{diam}(\Omega_j) \leq D_*$.*

Proof. The function u_j is a Neumann eigenfunction of $-\Delta$ at eigenvalue λ_j on the convex Ω_j . The Payne–Weinberger inequality gives $\lambda_j \geq \pi^2/\text{diam}(\Omega_j)^2$ for the first nonzero Neumann eigenvalue, but u_j is not necessarily a *first* eigenfunction. However, the number of Neumann eigenvalues below Λ on Ω_j is at most $C(\Lambda)|\Omega_j| = C(\Lambda)\pi$ by Weyl's asymptotics (uniform under area normalization for convex domains by Li–Yau bounds, see below). So λ_j is one of at most $C(\Lambda)\pi$ eigenvalues, and the smallest such eigenvalue dominates the others up to a multiplicative constant. Combined with Payne–Weinberger applied to the first eigenvalue, $\text{diam}(\Omega_j)$ is bounded by $D_*(\Lambda)$. $\square$

Combining the area normalization with an a priori bound on λ_j produces a compact family in the Hausdorff topology, modulo the question whether λ_j is bounded.

Remark 7 (Upper eigenvalue bound). *Schiffer's overdetermined problem is unrestricted in λ , so without further input λ_j is not bounded. Three natural ways to obtain $\lambda_j \leq \Lambda$:*

- *Restrict attention to counterexamples with λ below a chosen threshold Λ (e.g. to the first K eigenvalues). Deng's results already cover $K = 1, \dots, k_0$ for small k_0 on convex planar domains.*
- *Minimize λ over the class of counterexamples and bound the infimum by a comparison construction. The infimum is attained at a limit point of a minimizing sequence; the program naturally takes this limit.*
- *Bound λ_j in terms of the area-normalized fundamental Neumann gap by an isoperimetric inequality (a Fauconnier-type bound is conjectural here).*

The second option is the cleanest extremality choice and is the one adopted below.

Conjecture 8 (Schiffer infimum). *Define*

$$\Lambda_* := \inf\{\lambda \in (0, \infty) : \exists \Omega \subset \mathbb{R}^2 \text{ convex, non-disc, smooth, area } \pi, \text{ with a nonzero } u \text{ solving (1)}\}.$$

Then $\Lambda_ < \infty$.*

A simple comparison upper bound suffices: any ellipse close to a disc should produce *near-Schiffer* data with small Schiffer defect, which a perturbative bifurcation argument would either complete to a true Schiffer pair or rule out (Direction 3 territory). Assuming $\Lambda_* < \infty$, fix a minimizing sequence $\{(\Omega_j, u_j, \lambda_j)\}$ with $|\Omega_j| = \pi$, $\lambda_j \rightarrow \Lambda_*$, and apply Lemma 6 with $\Lambda = \Lambda_* + 1$ to obtain $\text{diam}(\Omega_j) \leq D_*$.

3 Step 2: spectral compactness via the 2412 toolkit

This is the step where arXiv:2412.06344 contributes most directly. de Dios Pont’s Step 4 (after Theorem 2.6) is: “By Li–Yau’s uniform Harnack inequality, the eigenfunctions should be equicontinuous, and thus have convergent subsequences in C^0 .” The cited references are [LY86] (Li–Yau, the parabolic kernel of the Schrödinger operator) and [Wan06] (Wang, gradient estimates). The relevant fact for us is:

Theorem 9 (Uniform gradient estimate, Wang). *Let $\Omega \subset \mathbb{R}^n$ be a smooth bounded convex domain and let u solve $-\Delta u = \lambda u$ in Ω with $\partial_\nu u = 0$ on $\partial\Omega$. Then*

$$\sup_{\Omega} |\nabla u|^2 \leq C_n \lambda \sup_{\Omega} u^2, \quad (2)$$

with C_n depending only on the dimension n .

The constant in (2) does *not* depend on the specific shape of Ω , only on dimension. This is the “uniform Harnack across convex domains” that the 2412 paper uses.

Corollary 10 (Equicontinuity of normalized Schiffer eigenfunctions). *Let $\{(\Omega_j, u_j, \lambda_j)\}$ be a sequence of smooth convex Schiffer counterexamples in $\mathbb{R}^2$ with $|\Omega_j| = \pi$, $\text{diam}(\Omega_j) \leq D_*$, and $\lambda_j \leq \Lambda_* + 1$. Normalize $\|u_j\|_{L^\infty(\Omega_j)} = 1$. Then $\{u_j\}$ is equicontinuous and uniformly bounded on the compact set $B_{D_*}(0) \supset \Omega_j$ (after zero-extension or a Whitney extension off Ω_j).*

Proof. $\sup |\nabla u_j| \leq \sqrt{C_2 \lambda_j} \leq \sqrt{C_2(\Lambda_* + 1)}$ by Theorem 9 applied with $n = 2$. The u_j are $C^{0,1}$ -bounded with constant $\sqrt{C_2(\Lambda_* + 1)}$ independent of j . Extend u_j by zero off Ω_j to obtain a sequence of $C^{0,1}$ -bounded functions on the compact set $\overline{B_{D_*}(0)}$, hence equicontinuous and uniformly bounded. $\square$

Corollary 11 (Spectral subsequential limit). *Under the hypotheses of Corollary 10, after extraction:*

$$\Omega_j \rightarrow \Omega_\infty \text{ in Hausdorff distance,} \quad u_j \rightarrow u_\infty \text{ uniformly on compacta,} \quad \lambda_j \rightarrow \lambda_\infty.$$

Ω_∞ is convex with $|\Omega_\infty| \leq \pi$ and $\text{diam}(\Omega_\infty) \leq D_*$. If Ω_∞ has nonempty interior, then $u_\infty \in C^{0,1}(\Omega_\infty)$ solves $-\Delta u_\infty = \lambda_\infty u_\infty$ in Ω_∞° .

Proof. Arzelà–Ascoli on $\{u_j\}$, Blaschke on $\{\Omega_j\}$, and bounded $\{\lambda_j\}$ all give subsequential limits. The PDE passes to the limit in distributions on every compact subset of Ω_∞° because of the uniform Lipschitz bound and uniform λ bound. $\square$

The remaining question is whether $|\Omega_\infty| > 0$, equivalently whether Ω_∞ is not collapsed to a segment. This is the *non-degeneracy* of the limit and is the main technical issue.

Lemma 12 (Non-degeneracy from a width lower bound). *If the inradius $r(\Omega_j) \geq r_* > 0$ uniformly, then Ω_∞ has nonempty interior.*

The hypothesis $r(\Omega_j) \geq r_*$ requires either a structural argument (e.g. thin domains have large Schiffer eigenvalue, contradicting $\lambda_j \leq \Lambda_* + 1$) or an explicit further normalization. The cleanest sufficient condition:

Conjecture 13 (Quantitative thin-strip exclusion). *There is a function $\omega : (0, \infty) \rightarrow (0, \infty)$ with $\omega(r) \rightarrow \infty$ as $r \rightarrow 0^+$ such that for every smooth convex Schiffer counterexample Ω in $\mathbb{R}^2$ with $|\Omega| = \pi$ and $r(\Omega) \leq r$, every Schiffer eigenvalue λ of Ω satisfies $\lambda \geq \omega(r)$.*

This is plausible by analogy with Faber–Krahn on thin domains: Dirichlet eigenvalues on a thin strip blow up like $(\text{width})^{-2}$. A Schiffer counterexample by definition has $u|_{\partial\Omega} = c$, and unless $c = 0$ this means the function does not vanish at the boundary, so the Dirichlet bound does not literally apply. A correct substitute is the Robin-type Faber–Krahn or a careful estimate on $u - c$, which solves $\Delta(u - c) + \lambda u = 0$, $\partial_\nu(u - c) = 0$, $(u - c)|_{\partial\Omega} = 0$. This is a mixed problem whose eigenvalues admit thin-strip lower bounds.

Conditional on Conjecture 13, the limit Ω_∞ has positive inradius and Step 2 closes.

4 Step 3: passing the overdetermined conditions

With $u_j \rightarrow u_\infty$ uniformly and $\Omega_j \rightarrow \Omega_\infty$ in Hausdorff distance, the Dirichlet trace is easy: $u_j|_{\partial\Omega_j} = c_j$. The constants c_j are bounded ($|c_j| \leq \sup|u_j| = 1$) and pass to $c_\infty \in [-1, 1]$. By uniform convergence and Hausdorff continuity of the boundary, $u_\infty|_{\partial\Omega_\infty} = c_\infty$.

The Neumann trace is harder. $\partial_\nu u_j = 0$ on $\partial\Omega_j$, and we want $\partial_\nu u_\infty = 0$ on $\partial\Omega_\infty$. This requires C^1 -convergence of u_j near the boundary and Hausdorff convergence of the normal vector fields, which fails at corners of Ω_∞ .

If $\partial\Omega_\infty$ is C^1 (which follows from strict convexity + Hausdorff convergence of strictly convex bodies on compact subsets of the boundary), then $\partial_\nu u_\infty = 0$ follows from interior gradient estimates and boundary $C^{1,\alpha}$ regularity for elliptic equations on convex domains (Krylov–Safonov / Caffarelli).

The two cases that fail are:

- $\partial\Omega_\infty$ has corners. Then Ω_∞ is a convex polygon (or has line-segment pieces), $\partial_\nu u_\infty$ is well-defined only away from the singularities, and the overdetermined condition holds in a weak sense.
- Ω_∞ has flat boundary pieces. Then $\partial\Omega_\infty$ contains line segments and the limit is no longer strictly convex.

Both can be handled by treating the singular part of $\partial\Omega_\infty$ as exceptional.

Proposition 14 (Limit Schiffer pair, conditional). *Conditional on Conjecture 13 and the absence of corners in $\partial\Omega_\infty$, the limit $(\Omega_\infty, u_\infty, \lambda_\infty)$ is a Schiffer pair: Ω_∞ is a smooth convex domain, $\lambda_\infty \geq \Lambda_*$, and u_∞ is a nontrivial Neumann eigenfunction of $-\Delta$ on Ω_∞ satisfying $u_\infty|_{\partial\Omega_\infty} = c_\infty$.*

By construction $\lambda_\infty = \Lambda_*$ (since the sequence is minimizing). Ω_∞ is therefore a Schiffer-minimizer: a convex domain of area π at which the Schiffer infimum is attained.

5 Step 4: dichotomy at the minimizer

There are two cases on the limit.

Case A: $\Omega_\infty = \text{disc}$. The unit disc satisfies Schiffer at each radial Neumann eigenvalue $\lambda_k^{\text{rad}} = (j'_{0,k})^2$. So $\Lambda_* \geq (j'_{0,1})^2 = (j_{1,1})^2$, and equality would mean $\Omega_\infty = B$ at the first radial Neumann eigenvalue. The sequence $\Omega_j \rightarrow B$ in Hausdorff distance with $\lambda_j \rightarrow (j_{1,1})^2$ and $u_j \rightarrow J_0(j_{1,1}r)$ (radial Bessel mode).

This case is excluded by a *local rigidity theorem near the disc*: there exists $\delta > 0$ such that no smooth convex non-disc Ω with $d_H(\Omega, B) < \delta$ admits a Schiffer eigenfunction. Existing perturbative work (Canuto, Kobayashi) gives such results for low Neumann eigenvalues. Whether the existing estimates cover the radial Schiffer eigenvalues λ_k^{rad} specifically is the technical question. A clean

version of Direction 3 (bifurcation from the disc) would either confirm this or provide an explicit non-disc branch.

Case B: $\Omega_\infty \neq \text{disc}$. Then Ω_∞ is a smooth convex non-disc Schiffer counterexample of area π at eigenvalue Λ_* . By minimality, Ω_∞ satisfies an Euler–Lagrange-type condition under area-preserving convex deformations. Specifically, every smooth one-parameter family of area-preserving deformations Ω_s with $\Omega_0 = \Omega_\infty$ that preserves Schiffer (and hence has $\lambda(\Omega_s)$ well-defined as a Schiffer eigenvalue) must satisfy $\partial_s \lambda(\Omega_s)|_{s=0} \geq 0$.

The shape-derivative formula on the disc-type domains is

$$\partial_s \lambda = - \int_{\partial\Omega} |\nabla u|^2 V_n dS,$$

where V_n is the normal velocity. On a Schiffer counterexample $|\nabla u|^2$ on $\partial\Omega$ is determined by the Cauchy data: $\partial_\nu u = 0$ and $u = c$ together with $-\Delta u = \lambda u$ at the boundary give $|\nabla u|^2|_{\partial\Omega} = (\partial_T u)^2 = 0$ (since $u|_{\partial\Omega} = c$ forces zero tangential derivative). Combined with $\partial_\nu u = 0$, the full gradient vanishes on $\partial\Omega$ and the shape derivative vanishes identically. So minimality in λ gives no information at first order.

At *second* order, the variational inequality reads (for a Schiffer-preserving deformation $V_n = f$ on $\partial\Omega$)

$$\partial_s^2 \lambda|_{s=0} \geq 0,$$

and the second shape derivative for a Schiffer pair has been computed (in different language) by Garofalo–Lewis, Sirakov, and others. The resulting nonnegativity is the right structural constraint, and it ought to be incompatible with $\Omega_\infty \neq \text{disc}$ when combined with the spectral pair structure.

6 What 2412 actually gives us, and what is still needed

1. Used directly from 2412.

- Theorem 9 (Wang/Li–Yau uniform gradient estimate). This is the key tool for Step 2. Equicontinuity of Schiffer eigenfunctions across the sequence is the precise analog of de Dios Pont’s Step 4.
- Smooth convex approximation (Lemma A.2 of 2412). This lets us replace non-smooth limits Ω_∞ by smooth approximations at every step without loss.

2. Borrowable from 2412 but not yet imported.

- The Brunn–Minkowski / log-concave inequalities used in 2412 to control the eigenvalue $\lambda_{\Omega,V}$ via weighted Faber–Krahn. The unweighted version of these inequalities is exactly what Conjecture 13 needs: thin convex domains have a quantitative Schiffer eigenvalue lower bound.
- The cap/lateral split (Section 2 of 2412). For Direction 2 this translates to a dichotomy between “elongated” and “compact” limits: an elongated Ω_∞ corresponds to the lateral collapse, a thin one to the cap collapse. The 2412 analysis says lateral collapse is the rigidity-friendly case.

3. Not in 2412, still needed.

- Local rigidity near the disc (Case A of Step 4). This belongs to Direction 3.

- The second-variation positivity at a minimizer (Case B). Needs an explicit Hessian-of-shape calculation for Schiffer pairs.
- Existence of a minimizing sequence with bounded eigenvalue (Conjecture 8). This requires constructing a near-Schiffer family with bounded λ , which is essentially a perturbative existence statement (Direction 3 again).

7 Concrete next deliverables

1. **Quantitative thin-strip exclusion** (Conjecture 13). Prove that thin convex domains have large Schiffer eigenvalues, by reducing to a Dirichlet eigenvalue estimate for $u-c$ via a Robin-type Faber–Krahn. This is a self-contained technical lemma; one page if it works.
2. **Apply Theorem 9 explicitly.** Write down the uniform Lipschitz bound for the Schiffer eigenfunction u on a convex domain in terms of λ and $|\Omega|$, and use it to verify equicontinuity in Corollary 10.
3. **Second shape derivative on a Schiffer pair.** Compute the Hessian of λ as a function of the boundary deformation, restricted to Schiffer-preserving deformations around a putative non-disc minimizer. Either show that the Hessian is sign-definite (ruling out Case B) or identify an explicit Schiffer-preserving direction that decreases λ , contradicting minimality.
4. **Verify the existing local rigidity near the disc.** Survey Canuto, Kobayashi, and any subsequent work to determine at which Neumann eigenvalues local rigidity is currently available, and at which eigenvalues it remains open. The radial Neumann eigenvalues $(j'_{0,k})^2$ are the relevant target.